Manav Rathi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.78
Eden International School	Class XII (CBSE)	2022	89.20%
Vivekananda Kendra Vidyalaya	Class X (CBSE)	2020	79.20%

WORK EXPERIENCE

AI Health Coach - Dietitian Portal | *Full-Stack Intern, 360 Technology*

- Independently developed a **Next.js Dietitian Portal** for managing patient cohorts in a CDC Prevent T2 program.

- Built real-time **patient dashboards and health analytics** integrating wearable metrics including steps, calories, and weight trends.

- Implemented **WebSocket-based chat**, **Zoom-integrated scheduling**, role-based authentication, route protection, and cohort management.

- Developed a production-grade UI using **TailwindCSS, reusable components, and TanStack React Query** for server-state management.

PROJECTS

Flappy Bird AI Agent using Q-Learning

- Built an autonomous Flappy Bird agent using Q-learning with state discretization and reward-based learning
- Engineered game mechanics using Pygame and visualized training performance using Matplotlib
- Achieved real-time gameplay decision-making by continuously updating Q-values based on agent's interactions

Chess Engine with Alpha-Beta Pruning

- Developed a functional chess engine supporting UCI protocol, move validation, and board evaluation
- Implemented Alpha-Beta pruning to optimize minimax search, enabling efficient decision-making at depth 3
- Designed modular components for board representation, move generation, and position evaluation logic

Image Caption Generator using CNN-LSTM

- Built an end-to-end image captioning system combining Xception CNN for feature extraction and LSTM for natural language generation
- Implemented data pre-processing pipelines for image embeddings, caption tokenization, and sequence padding to train on custom datasets
- Trained and evaluated the model with BLEU score metrics, enabling automatic caption generation for unseen images

CPU Scheduling Visualizer

- Simulated core CPU scheduling algorithms (FCFS, SRTF, Round Robin) with real-time Gantt chart visualization
- Parsed process data from external input files and implemented time-tracking, turnaround, and CPU utilization metrics
- Developed an interactive system mimicking OS-level process handling for educational and demonstration purposes

SKILLS

- **Programming Language :** C/C++, Python
- **Libraries/Framework** : NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
- **Fundamentals** : Data Structures & Algorithms, OS, OOPS, Machine Learning

- **Tools** : VS Code, GitHub, Google Colab, Adobe Premiere Pro

RELEVANT COURSES

- **Computer Science** : Introduction to Programming, Introduction to Artificial Intelligence, Data Structure and Algorithms
- **Mathematics** : Calculus-1, Calculus-2, Elementary Linear Algebra, Differential Equation, Introduction to Probability

SCHOLASTIC ACHIEVEMENTS

- **Harvard Project for Asian & International Relations** : 1 out of 100 globally invited by **Harvard** to attend the conference
- Solved **300+ DSA problems on LeetCode**, focusing on arrays, graphs, recursion, and dynamic programming
- Achieved **top 2% rank out of 1.1M+** candidates in **JEE Main**

POSITIONS OF RESPONSIBILITY

Placement Coordinator, Career Development Cell - IIT Hyderabad

- Initiated outreach to **50+ firms**, facilitating internship and placement opportunities for the 2027 cohort

Public Relations Core, Ek Bharat Shrestha Bharat (EBSB) - IIT Hyderabad

- Collaborated with **Ministry of Education** to lead campus-wide cultural exchange; ensured 24-hour report turnaround

Core Member, Rang De Manch (Theatre Club) - IIT Hyderabad

- Organized **Inter-IIT performances**; led production and team coordination, boosting club participation through rehearsals

EXTRACURRICULAR

- Represented **Rajasthan** at the **National Chess Championship** held in Bhopal, Madhya Pradesh
- Played a pivotal role in securing the 5th Position in Street Play, 7th in Stage Play at the **Inter-IIT Cultural Meet 6.0** held at **IIT Kharagpur**
- Participated in the **SBI Life Hack-AI-Thon** organized by Hack2skill, focusing on innovation, creativity, and collaboration in AI.